AI Interview Questions

Questions Only

1. Artificial Intelligence Fundamentals

1. What is Artificial Intelligence (AI)?

2. What are the major types of Artificial Intelligence?

3. What is the difference between narrow AI and general AI?

4. What are the main goals of Artificial Intelligence?

5. What is an intelligent agent?

6. What are the different types of intelligent agents?

7. What is the difference between AI, Machine Learning, and Deep Learning?

8. What is knowledge representation in AI?

9. What is reasoning in Artificial Intelligence?

10. What is the role of search algorithms in AI?

11. What is heuristic search?

12. What is the difference between supervised, unsupervised, and reinforcement learning?

2. Machine Learning

13. What is Machine Learning?

14. What are the main types of Machine Learning?

15. What is supervised learning?

16. What is unsupervised learning?

17. What is reinforcement learning?

18. What is the difference between classification and regression?

19. What is clustering?

20. What is the difference between training data and testing data?

21. Why do we split a dataset into training and testing sets?

22. What is validation data?

23. What is overfitting?

24. What is underfitting?

25. How can you prevent overfitting?

26. What is the bias-variance tradeoff?

27. What is cross-validation?

28. What is k-fold cross-validation?
29. What is feature engineering?
30. What is feature selection?
31. What is dimensionality reduction?
32. What is data preprocessing?
33. How do you handle missing values?
34. How do you handle categorical variables?
35. What is one-hot encoding?
36. What is label encoding?
37. What is normalization?
38. What is standardization?
39. What is data leakage?
40. What is class imbalance?
41. How can you handle an imbalanced dataset?
42. What is a baseline model?

3. Machine Learning Algorithms

43. How does Linear Regression work?
44. What are the assumptions of Linear Regression?
45. What is Multiple Linear Regression?
46. What is Logistic Regression?
47. Why is Logistic Regression used for classification?
48. What is the sigmoid function?
49. How does a Decision Tree work?
50. What are Gini Impurity and Entropy?
51. What is Information Gain?
52. What is Random Forest?
53. Why is Random Forest generally more robust than a single Decision Tree?
54. What is Gradient Boosting?
55. What is XGBoost?
56. What is K-Nearest Neighbors (KNN)?
57. How do you choose the value of K in KNN?
58. What is Support Vector Machine (SVM)?

59. What is the kernel trick in SVM?
60. What is Naive Bayes?
61. What is Principal Component Analysis (PCA)?
62. When would you use PCA?

4. Model Evaluation

63. What is accuracy?
64. What is precision?
65. What is recall?
66. What is F1-score?
67. What is a confusion matrix?
68. What is ROC-AUC?
69. When can accuracy be misleading?
70. What metrics are commonly used for regression?
71. What are MAE, MSE, and RMSE?
72. What is R-squared?
73. What is adjusted R-squared?
74. How do you select an evaluation metric for a Machine Learning problem?

5. Deep Learning

75. What is Deep Learning?
76. What is an Artificial Neural Network?
77. What is a neuron in a neural network?
78. What are input, hidden, and output layers?
79. What are weights and biases?
80. What is an activation function?
81. Why are activation functions required?
82. What is ReLU?
83. What is the difference between ReLU and Sigmoid?
84. What is backpropagation?
85. What is gradient descent?
86. What is the learning rate?
87. What is an epoch?

88. What is a batch?

89. What is batch size?

90. What is dropout?

91. What is batch normalization?

92. What is vanishing gradient?

93. What is exploding gradient?

94. What is a Convolutional Neural Network (CNN)?

95. What is a Recurrent Neural Network (RNN)?

96. What is LSTM?

97. What is GRU?

98. What is the difference between CNN and RNN?

6. Natural Language Processing

99. What is Natural Language Processing (NLP)?

100. What are the major applications of NLP?

101. What is tokenization?

102. What are stop words?

103. What is stemming?

104. What is lemmatization?

105. What is a Bag-of-Words model?

106. What is TF-IDF?

107. What are word embeddings?

108. What is Word2Vec?

109. What is the difference between CBOW and Skip-gram?

110. What is sentiment analysis?

111. What is Named Entity Recognition (NER)?

112. What is text classification?

113. What is the difference between traditional NLP and transformer-based NLP?

7. Generative AI and LLMs

114. What is Generative AI?

115. How is Generative AI different from traditional AI?

116. What is a Large Language Model (LLM)?

117. How does an LLM generate text?

118. What is a transformer architecture?

119. What is self-attention?

120. What is multi-head attention?

121. What are tokens in an LLM?

122. What is tokenization in LLMs?

123. What is an embedding?

124. What is a context window?

125. What is temperature in text generation?

126. What are top-k and top-p sampling?

127. What is prompt engineering?

128. What is zero-shot prompting?

129. What is few-shot prompting?

130. What is hallucination in LLMs?

131. How can LLM hallucinations be reduced?

132. What is fine-tuning?

133. What is the difference between fine-tuning and prompt engineering?

8. Retrieval-Augmented Generation (RAG)

134. What is Retrieval-Augmented Generation (RAG)?

135. Why is RAG used with Large Language Models?

136. What are the main components of a RAG pipeline?

137. What is document ingestion?

138. What is document chunking?

139. Why is chunking important in RAG?

140. What are embeddings in a RAG system?

141. What is semantic search?

142. What is a vector database?

143. What is FAISS?

144. How does vector similarity search work?

145. What is cosine similarity?

146. What is the difference between keyword search and semantic search?

147. What is a retriever?

148. What is a reranker?

149. What is the difference between RAG and fine-tuning?

150. How can you improve the retrieval quality of a RAG system?

151. How do you evaluate a RAG application?

152. What are common failure cases in RAG systems?

153. How would you build a multi-document RAG chatbot?

9. Computer Vision

154. What is Computer Vision?

155. What are the major applications of Computer Vision?

156. What is image classification?

157. What is object detection?

158. What is image segmentation?

159. What is the difference between image classification and object detection?

160. What is a convolution operation?

161. What is a kernel or filter in CNNs?

162. What is pooling?

163. What is data augmentation?

164. What is transfer learning?

165. Why is transfer learning useful for image classification?

166. What is image preprocessing?

167. What is the role of OpenCV in Computer Vision?

10. AI/ML Project-Based Questions

168. Explain your most important AI/ML project.

169. What problem were you trying to solve in your project?

170. Why did you choose Machine Learning for the problem?

171. How did you collect and prepare the dataset?

172. What preprocessing steps did you perform?

173. How did you select the features?

174. Which Machine Learning algorithms did you try?

175. Why did you choose the final model?

176. How did you evaluate your model?

177. What challenges did you face during the project?

178. How did you handle missing or noisy data?

179. How did you handle class imbalance?

180. How did you improve the model performance?

181. How did you deploy your Machine Learning model?

182. How would you scale your project for production?

183. What would you improve if you had more time?

11. Scenario-Based AI Questions

184. How would you build a Machine Learning model for fraud detection?

185. How would you design an AI system for customer churn prediction?

186. How would you build a recommendation system?

187. How would you detect data drift in a production ML system?

188. What would you do if your model performs well on training data but poorly on test data?

189. What would you do if your dataset contains millions of records?

190. How would you handle a dataset with highly imbalanced classes?

191. How would you choose between two models with similar accuracy?

192. How would you reduce inference latency in an AI application?

193. How would you monitor a deployed ML model?

194. How would you design a chatbot using an LLM?

195. How would you build a RAG system for company documents?

196. How would you reduce hallucinations in an enterprise chatbot?

197. How would you protect sensitive information in an AI application?

198. How would you evaluate whether an AI system is ready for production?